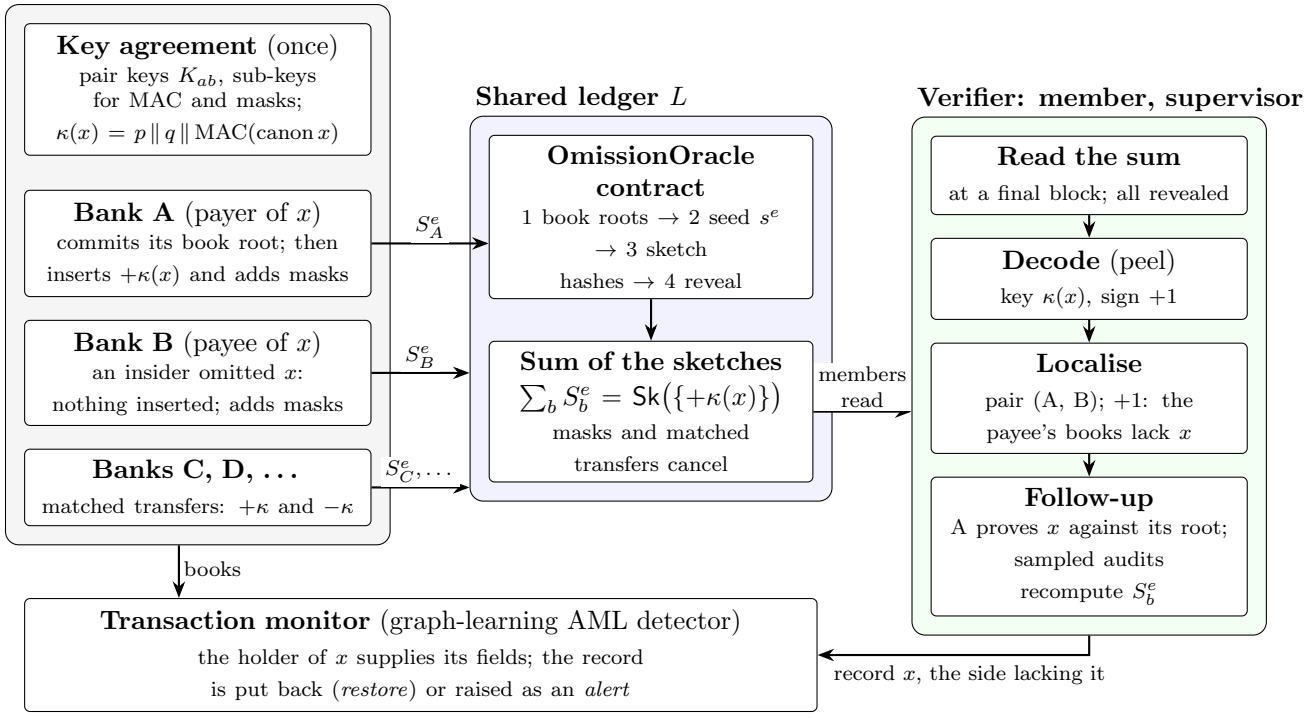

Members: books stay with each bank
every epoch e (e.g. hourly)
Key agreement (once)
pair keys K_{ab}, sub-keys for MAC and masks;
$\kappa(x) = p \,\|\, q \,\|\, \mathrm{MAC}(\mathrm{canon}\, x)$
Bank A (payer of x)
commits its book root; then inserts $+\kappa(x)$ and adds masks
Bank B (payee of x)
an insider omitted x: nothing inserted; adds masks
Banks C, D, ...
matched transfers: $+\kappa$ and $-\kappa$
S_A^e
S_B^e
$S_C^e, \dots$
Shared ledger L
OmissionOracle contract
1 book roots → 2 seed s^e
→ 3 sketch
hashes → 4 reveal
Sum of the sketches
$\sum_b S_b^e = \mathsf{Sk}(\{+\kappa(x)\})$
masks and matched transfers cancel
members read
Verifier: member, supervisor
Read the sum
at a final block; all revealed
Decode (peel)
key $\kappa(x)$, sign +1
Localise
pair (A, B); +1: the payee's books lack x
Follow-up
A proves x against its root; sampled audits recompute S_b^e
books
Transaction monitor (graph-learning AML detector)
the holder of x supplies its fields; the record is put back (restore) or raised as an alert
record x, the side lacking it